

Laxmi Charitable Trust's  
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce  
**Department of Information Technology (B.Sc.I.T Semester V)**  
Full Stack development MERN Practical

M2 Practical-8

|                               |                                    |
|-------------------------------|------------------------------------|
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| Class:TYIT                    | Batch:1                            |
| Date of Assignment: 26-9-2026 | Date/Time of Submission: 26-9-2026 |

Project Title: College Canteen Pre-Order System  
Develop a web-based College Canteen Pre-Order System.  
It should allow students to:

1. View the available food items and their prices.
2. Enter student details such as name, roll number, and course.
3. Select a food item and quantity.
4. Place an order using a POST API.
5. Store the order details in MongoDB.
6. Retrieve the stored orders using a GET API.
7. Display the available food items and placed orders on the React page.

APIs to be implemented

- GET /food – Retrieve food items.
- POST /orders – Add a new food order.
- GET /orders – Retrieve placed orders.

Technologies

- React
- Axios
- Node.js
- Express.js
- MongoDB

Expected Output

The final application should contain:

- Food Item List
- Student Order Form
- Place Order button
- Order List/Table

Design the interface and decide the appropriate fields, MongoDB document structure, and API flow.

**Code:-**

**App.jsx**

```
import { useEffect, useState } from "react";
import axios from "axios";
import "./App.css";

function App() {
  const [food, setFood] = useState([]);
  const [orders, setOrders] = useState([]);
```

```
const [form, setForm] = useState({
  studentName: "",
  rollNo: "",
  course: "",
  foodItem: "",
  quantity: 1
});

// Load food and orders
useEffect(() => {
  axios
    .get("http://localhost:5000/food")
    .then((res) => setFood(res.data))
    .catch((err) => console.log(err));

  getOrders();
}, []);

// Get orders
const getOrders = () => {
  axios
    .get("http://localhost:5000/orders")
    .then((res) => setOrders(res.data))
    .catch((err) => console.log(err));
};

// Input change
const change = (e) => {
  setForm({
    ...form,
    [e.target.name]: e.target.value
  });
};

// Place order
const order = (e) => {
  e.preventDefault();

  axios
    .post("http://localhost:5000/orders", {
```

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```
...form,
rollNo: Number(form.rollNo),
quantity: Number(form.quantity)
})
.then(() => {
  alert("Order Placed!");

  getOrders();

  setForm({
    studentName: "",
    rollNo: "",
    course: "",
    foodItem: "",
    quantity: 1
  });
})
.catch((err) => {
  console.log(err);
  alert("Order failed");
});
});

return (
  <div className="container">

    <h1>College Canteen Pre-Order System</h1>

    <h2>Food Items</h2>

    <div className="food">
      {food.map((f) => (
        <div className="food-card" key={f.name}>
          <b>{f.name}</b>
          <p>₹ {f.price}</p>
        </div>
      ))}
    </div>

    <h2>Student Order Form</h2>

    <form onSubmit={order}>

      <input
        name="studentName"
        placeholder="Student Name"
        value={form.studentName}
        onChange={change}
        required
      />

      <input
        type="number"
        name="rollNo"
```

```
placeholder="Roll No"
value={form.rollNo}
onChange={change}
required
/>

<input
  name="course"
  placeholder="Course"
  value={form.course}
  onChange={change}
  required
/>

<select
  name="foodItem"
  value={form.foodItem}
  onChange={change}
  required
>
  <option value="">Select Food</option>

  {food.map((f) => (
    <option key={f.name} value={f.name}>
      {f.name} - ₹ {f.price}
    </option>
  ))}
</select>

<input
  type="number"
  name="quantity"
  min="1"
  value={form.quantity}
  onChange={change}
  required
/>

<button type="submit">
  Place Order
</button>

</form>

<h2>Placed Orders</h2>

<table>
  <thead>
    <tr>
      <th>Name</th>
      <th>Roll No</th>
      <th>Course</th>
      <th>Food</th>
      <th>Price</th>
```

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```
        <th>Qty</th>
        <th>Total</th>
      </tr>
    </thead>

    <tbody>
      {orders.map((o) => (
        <tr key={o._id}>
          <td>{o.studentName}</td>
          <td>{o.rollNo}</td>
          <td>{o.course}</td>
          <td>{o.foodItem}</td>
          <td>₹ {o.price}</td>
          <td>{o.quantity}</td>
          <td>₹ {o.total}</td>
        </tr>
      ))}
    </tbody>
  </table>

</div>
);
}
```

export default App;

### App.css:-

```
body {
  font-family: Arial, sans-serif;
  background: #f2f2f2;
  margin: 0;
}

.container {
  width: 90%;
  margin: auto;
  padding: 20px;
}

h1 {
  text-align: center;
  color: #333;
  margin-bottom: 30px;
}

h2 {
  color: #333;
  margin-top: 25px;
}

.food {
  display: flex;
```

```
  gap: 15px;
  flex-wrap: wrap;
}

.food-card {
  background: white;
  padding: 15px;
  width: 150px;
  border-radius: 8px;
  text-align: center;
  box-shadow: 0 2px 5px #ccc;
}

.food-card b {
  font-size: 18px;
}

.food-card p {
  margin-bottom: 0;
}

form {
  background: white;
  padding: 20px;
  margin: 20px 0;
  border-radius: 8px;
}

input,
select,
button {
  display: block;
  width: 100%;
  padding: 10px;
  margin: 8px 0;
  box-sizing: border-box;
}

input,
select {
  border: 1px solid #ccc;
  border-radius: 4px;
}

button {
  background: green;
  color: white;
  border: none;
  cursor: pointer;
  border-radius: 4px;
}

button:hover {
  background: darkgreen;
```

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```
}

table {
  width: 100%;
  background: white;
  border-collapse: collapse;
  margin-top: 20px;
}

th,
td {
  padding: 10px;
  border: 1px solid #ccc;
  text-align: center;
}

th {
  background: #333;
  color: white;
}

td {
  color: #555;
}

Service.js:-

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();

app.use(cors());
app.use(express.json());

// MongoDB Connection
mongoose
  .connect("mongodb://127.0.0.1:27017/college")
  .then(() => {
    console.log("MongoDB Connected");
    seedOrders();
  })
  .catch((err) => {
    console.log("MongoDB Error:", err);
  });

// Order Schema
const Order = mongoose.model(
  "students",
  new mongoose.Schema({
    studentName: String,
    rollNo: Number,
    course: String,
```

```
    foodItem: String,
    price: Number,
    quantity: Number,
    total: Number
  })
);

// Food Items
const food = [
  { name: "Pizza", price: 100 },
  { name: "Veg Burger", price: 60 },
  { name: "Sandwich", price: 50 },
  { name: "Samosa", price: 20 },
  { name: "Cold Coffee", price: 70 }
];

// Default Orders
const defaultOrders = [
  {
    studentName: "Riya",
    rollNo: 5,
    course: "B.Sc. IT",
    foodItem: "Pizza",
    price: 100,
    quantity: 2,
    total: 200
  },
  {
    studentName: "Priya",
    rollNo: 8,
    course: "B.Sc. IT",
    foodItem: "Veg Burger",
    price: 60,
    quantity: 1,
    total: 60
  },
  {
    studentName: "Aashu",
    rollNo: 12,
    course: "B.Sc. IT",
    foodItem: "Sandwich",
    price: 50,
    quantity: 3,
    total: 150
  },
  {
    studentName: "Sneha",
    rollNo: 15,
    course: "B.Sc. IT",
    foodItem: "Samosa",
    price: 20,
    quantity: 4,
    total: 80
  },
];
```

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```
{
  studentName: "Nitesh",
  rollNo: 7,
  course: "CS",
  foodItem: "Cold Coffee",
  price: 70,
  quantity: 2,
  total: 140
}
];

// Insert default orders only if collection is empty
async function seedOrders() {
  const count = await Order.countDocuments();

  if (count === 0) {
    await Order.insertMany(defaultOrders);
    console.log("Default orders inserted");
  }
}

// GET /food
app.get("/food", (req, res) => {
  res.json(food);
});

// POST /orders
app.post("/orders", async (req, res) => {
  try {
    const {
      studentName,
      rollNo,
      course,
      foodItem,
      quantity
    } = req.body;

    const item = food.find((f) => f.name === foodItem);

    if (!item) {
      return res.status(400).json({
        message: "Food item not found"
      });
    }

    const total = item.price * quantity;

    const order = new Order({
      studentName,
      rollNo,
      course,
      foodItem,
      price: item.price,
      quantity,
```

```
total
    });
  });

  await order.save();
  res.json(order);
} catch (error) {
  res.status(500).json({
    message: "Order failed"
  });
}
});

// GET /orders
app.get("/orders", async (req, res) => {
  try {
    const orders = await Order.find();
    res.json(orders);
  } catch (error) {
    res.status(500).json({
      message: "Unable to fetch orders"
    });
  }
});

// Server
app.listen(5000, () => {
  console.log("Server running on 5000");
});
```

### Output:-



**College Canteen Pre-Order System**

Food Items:

Pizza ₹120 Veg Burger ₹80 Sandwich ₹50 Samosa ₹20 Cold Coffee ₹35

Student Order Form

Placed Orders

| Name        | Roll No | Course  | Food        | Price | Qty | Total |
|-------------|---------|---------|-------------|-------|-----|-------|
| Piya        | 5       | B.Sc IT | Pizza       | ₹120  | 2   | ₹240  |
| Piya        | 3       | B.Sc IT | Veg Burger  | ₹80   | 1   | ₹80   |
| Aashu       | 12      | B.Sc IT | Sandwich    | ₹50   | 2   | ₹100  |
| Grisha      | 18      | B.Sc IT | Samosa      | ₹20   | 4   | ₹80   |
| Nitesh      | 7       | CS      | Cold Coffee | ₹35   | 2   | ₹70   |
| Aashu Yadav | 30      | B.A B   | Sandwich    | ₹50   | 1   | ₹50   |